SECTION-II MACROECONOMICS THEORY

CHAPTER 8

THEORY OF INCOME AND EMPLOYMENT

Course of Action
Maximum Time = 3:15 Hrs
Maximum Questions = 90

Learning Objectives

The students will learn:

- *Meaning of Aggregate Demand and its components.*
- *The concept of propensity to consume and save and how to calculate them.*
- *The short – run equilibrium that the economy can achieve through both AD-AS approach and I-S approach.*
- *What happens during the times of disequilibrium in the economy.*
- *The concept of Investment Multiplier and its relationship with propensity to consume, save, investment and income, and the ways to calculate them.*
- *The causes of deflation and inflation in the economy and how can they be corrected through various fiscal and monetary measures.*

Topic-1
Basic concepts of Aggregate Demand and Aggregate Supply **Page No. 150**

Topic-2
Short-run Equilibrium and Concept of Multiplier **Page No. 157**

Topic-3
Concepts of Excess Demand and Deficient Demand **Page No. 164**

Topic-1 Basic concepts of Aggregate Demand and Aggregate Supply

Concepts covered: Aggregate demand, Aggregate supply, Propensity to Consume, Propensity to Save, Consumption Function, Saving Function, Investment Function, Marginal Efficiency of Capital

Revision Notes

- **Aggregate Demand :** Aggregate Demand refers to the total demand for all goods and services in the economic system as a whole. This is expressed in terms of total expenditure made in the economy.
- **Constituents of Aggregate Demand:** In an open economy constituents of aggregate demand. are :

(i) Consumption Demand
(ii) Investment Demand
(iii) Government Expenditure
(iv) Net Exports

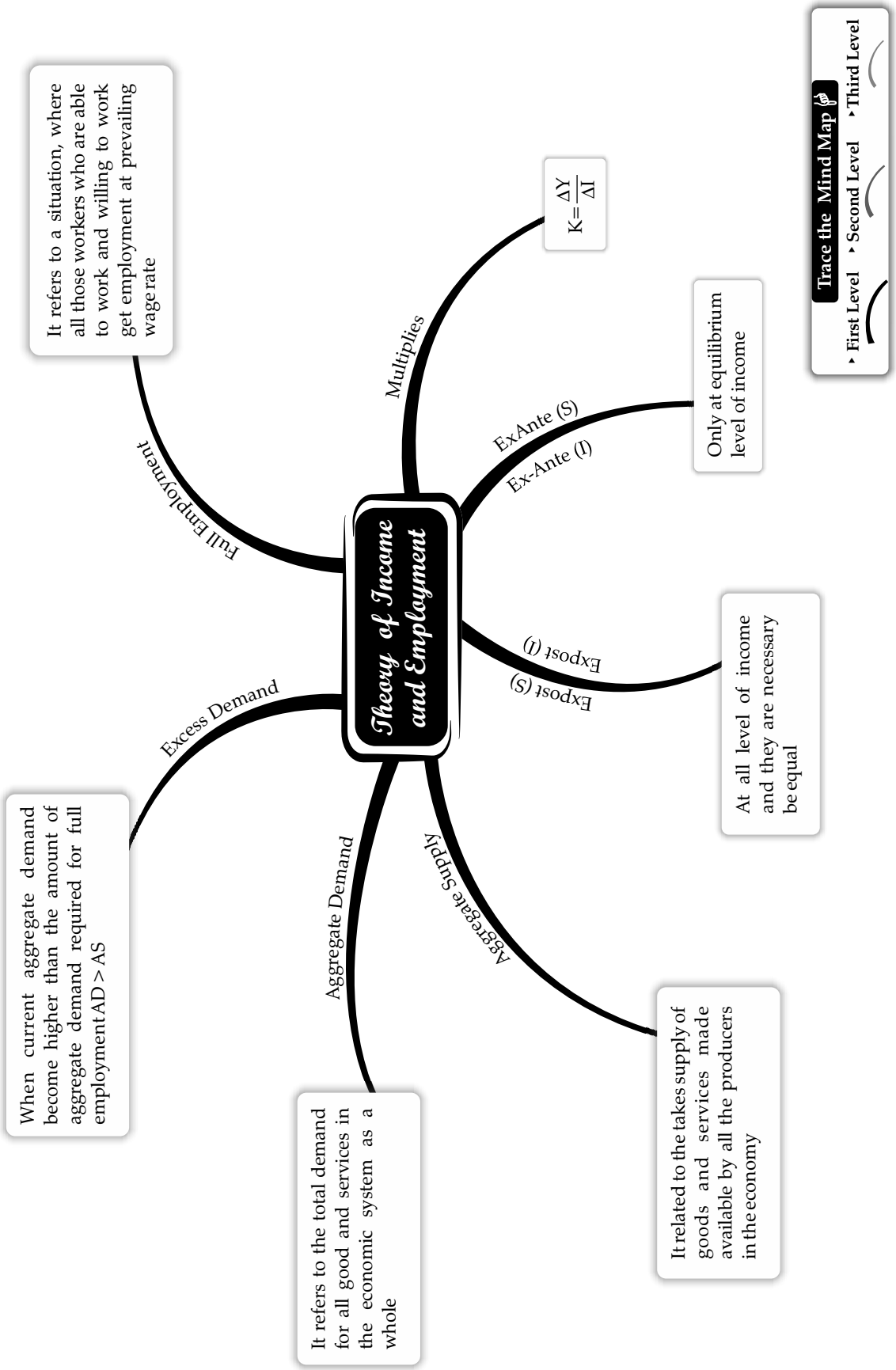
$\therefore AD = C + I + G + (X - M)$

Scan to know more about this topic
Aggregate Demand and aggregate supply

- **Aggregate Supply :** The concept of aggregate supply is related to the total supply of goods and services made available by all the producers in the economy. It can be expressed in three forms :

(i) Money value of goods and services produced during a year in an economy, i.e., National Income.
(ii) In the form of total income, i.e., consumption + saving.
(iii) In the form of minimum income which the firm will receive as sale proceeds from the sale of goods and services.

- **Aggregate Supply Price :** The minimum expected sale proceeds of the output resulting from a given amount of employment is called the 'Aggregate Supply Price' of that output.
- **Aggregate Supply Schedule :** This schedule represents the addition of consumption expenditure and saving at various levels of income.


Theory of Income and Employment
Full Employment
It refers to a situation, where all those workers who are able to work and willing to work get employment at prevailing wage rate
Multiplies
K= ΔY/ΔI
Ex-Ante (S)
Ex-Ante (I)
Only at equilibrium level of income
Expost (S)
Expost (I)
At all level of income and they are necessary be equal
Aggregate Supply
It related to the takes supply of goods and services made available by all the producers in the economy
Aggregate Demand
It refers to the total demand for all good and services in the economic system as a whole
Excess Demand
When current aggregate demand become higher than the amount of aggregate demand required for full employment AD > AS
Trace the Mind Map
First Level
Second Level
Third Level

- **Propensity to Consume :** It expresses the consumption levels at different levels of income.
- **Average Propensity to Consume (APC) :** It is the ratio of consumption expenditure to any particular level of income.

$$APC = \frac{\text{Consumption } (C)}{\text{Income } (Y)}$$

- **Marginal Propensity to Consume (MPC) :** It is the ratio of change in consumption to change in income.

$$MPC = \frac{\text{Change in Consumption } (C)}{\text{Change in Income } (Y)} = \frac{\Delta C}{\Delta Y}$$

- **Propensity to Save :** It is the ratio of saving to income at different levels of income.
- **Average Propensity to Save (APS) :** It is the ratio of saving to income.

$$APS = \frac{\text{Saving } (S)}{\text{Income } (Y)}$$

- **Marginal Propensity to Save (MPS) :** It is the ratio of change in saving to a change in income.

$$MPS = \frac{\text{Change in Saving}}{\text{Change in Income}} = \frac{\Delta S}{\Delta Y}$$

- **Relationship between Propensity to Save and Propensity to Consume :**

$$MPC + MPS = 1$$
or $$MPC = 1 - MPS$$
or $$MPS = 1 - MPC$$
$$APC + APS = 1$$
or $$APC = 1 - APS$$
or $$APS = 1 - APC$$

- **Consumption Function :** The functional relationship between consumption and income is called as Consumption Function.

$$C = f(Y)$$

The algebraic expression of consumption function is given by :

$$C = \bar{C} + b(Y)$$

where, C = consumption

$\bar{C}$ = Minimum level of Consumption at zero Income.
b = Marginal Propensity to Consume
Y = Income

- **Saving Function :** It denotes the relation between saving and income. It shows the desire of savings at various levels of income.

$$S = f(Y).$$

The algebraic expression of saving function is given by:

$$S = (-)\ \bar{S} + b(Y)$$

Where, S = Saving

$\bar{S}$ = Level of saving when Income is Zero.
b = Marginal Propensity to Save
Y = Income

Example:

Complete the following table:

Income (Y) ₹	Consumption (C)	APS	MPS	APC	MPC
0	50	---	---	---	---
100	70	---	---	---	---
200	100	---	---	---	---
300	175	---	---	---	---

Ans.

Income (Y) ₹	Consumption (C)	Change in Income (ΔY)	Change in Consumption (ΔC)	APS = 1 - APC	MPS = 1 - MPC	APC $= \frac{C}{Y}$	MPC $= \frac{\Delta C}{\Delta Y}$
0	50	-	-	-	-	-	-
100	70	100	20	0.3	0.8	0.7	0.2
200	100	100	30	0.5	0.7	0.5	0.3
300	175	100	75	0.42	0.25	0.58	0.75

- **Investment Function :** It refers to the behaviour of investment corresponding to different levels of income / employment.
- **Induced Investment :** It depends upon income and profit in the economy. Investment made with expectation of profit is called induced investment. It depends upon (i) Marginal efficiency of capital, and (ii) rate of interest.
- **Autonomous Investment :** This investment is independent of income and employment. Such investment is made by the government with the motive of social welfare.
- **Marginal Efficiency of Capital (MEC) :** Marginal efficiency of capital is the expected rate of return on an additional unit of capital goods over its cost.

$$MEC = \frac{\text{Prospective Yield}}{\text{Cost (Supply Price of Capital)}}$$

Key Term

- **Investment :** Investment expenditure includes expenditure for producer's durable equipment, new construction and the change in inventories.

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. Complete the following table: **[ISC 2023]**

INCOME (Y)	SAVING (S)	APC
0	(–)12	
20	6	

Ans.

INCOME (Y)	SAVING (S)	Consumption (C) = Y – S	APC = $\frac{C}{Y}$
0	(–)12	12	$\frac{12}{0} = \infty$
20	6	14	$\frac{14}{20} = 0.7$

Q.2. What is meant by MPS? How is MPS related to investment multiplier? **[ISC 2020]**

Ans. Marginal propensity to save It is the ratio of change in saving to change in income.

MPS = $\frac{\Delta S}{\Delta Y}$ MPS and investment multiplier are inversely related to each other.

K = $\frac{1}{1-MPC}$ or $\frac{1}{MPS}$

[ISC Marking Scheme, 2020]

Examiner's Comment

- Majority of the candidates got confused between the definitions of MPS and APS. Most candidates defined MPS as change in saving for change in income. This was a very vague definition of MPS and had nothing to do with its concept. The answer expected from the candidates was that MPS is the ratio of change in saving to change in income.

Answering Tip

- Train students to write the full form of MPS, its definition and its formula even if not asked for in the question. Explain that MPS is the ratio of change in savings to change in income. They can explain the relationship between MPS and investment multiplier through K = 1/1 – MPC or 1/MPS.

Q.3. Give the meaning of aggregate supply. **[OEB]**

Q.4. Define ex-ante consumption. **[OEB]**

Ans. Ex-ante consumption can be defined as the planned consumption made by all the consumers in the economy.

Q.5. Give any one reason why does the consumption curve does not start from the origin? **[OEB]**

Ans. The consumption curve does not start from the origin because even at zero level of income there is some consumption that needs to be fulfilled. Hence, Consumption curve does not start from the origin.

Q.6. State giving reason whether the following statement is true or false: **[OEB]**

"The sum of MPC and MPS is equal to 2."

Ans. False

Reason: As the sum of MPC and MPS is equal to one.

Q.7. "In a two-sector model aggregate demand comprises of consumption expenditure and government expenditure." **[OEB]**

Ans. False

Reason: This is because in a two-sector model aggregate demand comprises of consumption expenditure and investment expenditure.

Q.8. "The sum of APC and APS is equal to 2." **[OEB]**

Ans. False

Reason: This is because sum of Average propensity to consume and average propensity to save is equal to one.

2 Marks Questions

Q.1. What is meant by propensity to consume? **[ISC 2015]**

Ans. Propensity to Consume refers to different possible amounts of consumption expenditure which the consumers desire to spend at different levels of national income. **[ISC Marking Scheme 2015]**

Examiner's Comment

- In some cases, 'propensity to consume' was explained as a micro concept i.e. consumption expenditure of a consumer which is incorrect.

This question is for practice and its solution is available at the end of the chapter

It was written as $C = f_{(Y)}$ instead of defining it with respect to different levels of National Income.

Answering Tip

- The meaning of propensity should be explained with reference to consumption expenditure at different levels of national income.

Q.2. Give difference between autonomous consumption and induced consumption? [OEB]

Ans. Autonomous consumption can be defined as the consumption made at zero level of income. It is that consumption which is necessary for survival when earnings or income is zero. Induced consumption can be defined as that consumption which rises with income level.

Q.3. Explain the significance of 45° line in Keynesian economics? [OEB]

Ans. The significance of 45° line in Keynesian economics is that aggregate supply curve is upward sloping straight line starting from the origin because Income= consumption + saving and income is measured on x-axis and consumption and saving are measured on y-axis.

Q. 4. Which of the following cannot have negative value? Give reasons. [CBSE OD Set-III, Comptt., 2015]

(i) Average Propensity to Save

(ii) Marginal Propensity to Save

Ans. (i) APS can have a negative value because at very low level of income there is dis-saving.

(ii) MPS cannot have a negative value as it is the ratio of DS and DY.

Q.5. What do you mean by propensity to save? [OEB]

3 Marks Questions

Q.1. Briefly discuss components of Aggregate Demand in an economy. [ISC 2020]

OR

Discuss the main components of aggregate demand in an economy with the help of a diagram. [ISC 2016]

Ans. In an open economy, constituents of AD are:

(i) Consumption Expenditure

(ii) Investment Expenditure

(iii) Government Expenditure

(iv) Net Exports

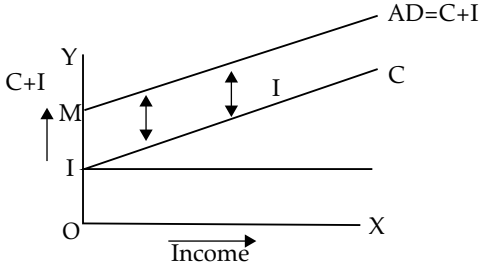
AD = C+ I + G + (X – M)

(i) Private Consumption Expenditure: The total demand for all goods and services by the household in an economy during an accounting year is termed as Private Consumption Expenditure. It is determined by the level of personal disposable income of the economy.

(ii) Private Investment Expenditure: The expenditure of households and private investor to purchase goods or services that adds to their stock of capital is termed as Private Investment Expenditure. It mainly depends on market rate of interest.

(iii) Government Expenditure: Government Expenditure includes the total expenditure of government on the purchase of consumption goods and investment expenditure. There is a significant difference between government and private investment. Private investments are done on consideration of profit and termed as Induced Investment. Government investment expenditure done on considerations of social welfare like construction of roads, schools, dams and flyover are termed as Autonomous Investment.

(iv) Net Exports: Net exports is the difference between the goods exported to the foreign country and the goods imported from the foreign country by the households and private investors or government of the country.



Examiner's Comment

- Many candidates did not write the definition of aggregate demand. Some candidates only wrote the two-sector model although they were supposed to write all the components.

Answering Tip

- Teach students to start their answers with a definition, to write the formula with all the components as no marks are awarded if only a specific sector is written about. Ensure that the students explain the components properly.

Q.2. What is meant by average propensity to consume? Explain its relationship with average propensity to save. [ISC 2019]

OR

Show the relationship between APC and APS. [ISC 2016]

Ans. Average propensity to consume.

APC refers to the proportion of income desired to be spent on consumption.

$\therefore$ APC = $\frac{Y}{C}$

This question is for practice and its solution is available at the end of the chapter

Relationship between APC and APS.

$Y = C + S$

Or, $\frac{Y}{Y} = \frac{C}{Y} + \frac{S}{Y}$ (Dividing both the sides by Y)

Or, $1 = APC + APS$

$\therefore$ $APC = 1 - APS$

$\therefore$ $APS = 1 - APC$

When APC increases, APS decreases

$\therefore$ It shows negative correlation.

[ISC Marking Scheme, 2019]

Examiner's Comment

- Several candidates defined APC but did not write its formula. Candidates wrote the relationship of APC with APS directly without deriving it.

Answering Tip

- Learn the derivation of the relationship between APC and APS with all the necessary steps.

Q.3. Complete the following table: **[ISC 2018]**

Income (Y) ₹	Consumption (C)	APS	MPS
0	40	---	---
50	70	---	---
100	100	---	---
150	120	---	---

Ans.

Income (Y) ₹	Consumption (C)	Savings (S) = Y – C	Change in Savings (ΔS)	Change in Income (ΔY)	APS = $\frac{S}{Y}$	MPS = $\frac{\Delta S}{\Delta Y}$
0	40	– 40	–	–	–	–
50	70	– 20	20	50	– 0.4	0.4
100	100	0	20	50	0	0.4
150	120	30	30	50	0.2	0.6

Examiner's Comment

- Many candidates did not write the missing value of APS in the table. A few candidates wrote the value of MPC instead of the value of MPS.

Answering Tip

- Teach the calculation of the values of APS and MPS with the formulae.

Q.4. Define marginal propensity to consume. How is it different from marginal propensity to save? **[ISC 2017]**

Ans. Marginal propensity to consume is the ratio of the change in consumption to the change in income. To calculate it, we divide change in consumption by a change in income.

$$MPC = \frac{\Delta C}{\Delta Y}$$

Here MPC = Marginal propensity to consume

ΔC = change in consumption

ΔY = change in income

MPS is the ratio of the change in saving to the change in income.

$$MPS = \frac{\Delta S}{\Delta Y}$$

[ISC Marking Scheme, 2017]

Examiner's Comment

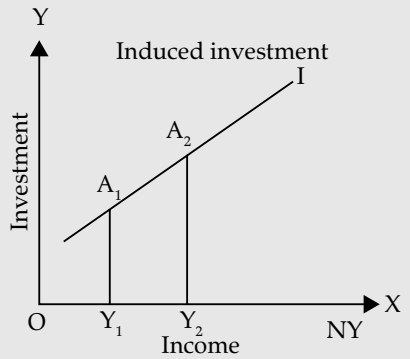
- A number of candidates did not write that MPC is a ratio of 'change in consumption' to 'change in income'. The word 'change' was missing in many answers; same was the case with the MPS.

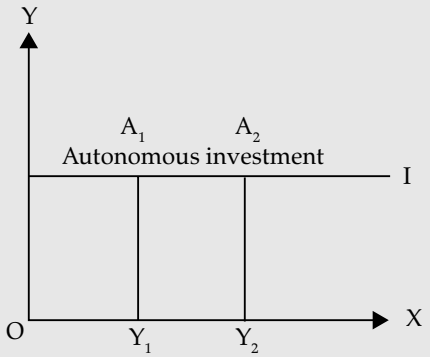
Answering Tip

- Tell students to write the word 'change' in both MPC and MPS. The expressions MPC = ΔC/ΔY MPS = ΔS/ΔY should be written.

Q.5. Explain the difference between induced investment and autonomous investment. **[ISC 2016]**

Ans. The difference between induced and autonomous investment. Autonomous investment is the one that is not affected by the change in the level of income or output. Therefore, it is income inelastic and may change because of non-income factors. Induced investment is the one which is undertaken as a result of change in the level of income. It varies directly with the change in income.





[ISC Marking Scheme, 2016]

Examiner's Comment

- Induced and autonomous investments were defined in an erratic manner by many candidates, without considering income. Some candidates gave the definition correctly but did not give the diagram.

Answering Tip

- Students should be made to understand how investment is related to income. Use diagrams to explain the difference.

Q.6. What is meant by autonomous consumption? Explain with the help of a diagram. [ISC 2015]

4 Marks Questions

Q.1. Read the passage given below and answer the following questions. [ISC Semester-I, 2022]

The Great Depression of 1930s proved that without the government's intervention, the economy cannot create employment opportunities. In India, MGNREGA programmes are based on the Keynesian aggregate demand strategy of direct job creation by the government. MGNREGA increases the purchasing power of those individuals whose MPC is the highest.

(i) What is meant by MPC?

(ii) State the components of aggregate demand.

(iii) How is MPS obtained from MPC?

Ans. (i) MPC is Marginal Propensity to Consume. The ratio of change in consumption (C) to change in income (Y) is known as marginal propensity to consume. MPC indicates the proportion of additional income that is being spent on consumption.

(ii) Aggregate demand is made up of four components: consumption, investment, government spending and net exports (exports–imports). These are discussed below:

(a) Household (or Private) Consumption Demand (C)

(b) Private Investment Demand (I)

(c) Government Demand for Goods and Services (G)

(d) Net Exports (Exports - Imports) Demand

(iii) The sum of MPC and MPS is equal to one. Thus, MPS = 1 – MPC.

Q.2. If MPC is 0.5 and initial consumption in the economy is rupees ₹200 crores. Calculate the equilibrium level of income by assuming autonomous investment is of ₹160 crores. [OEB]

Ans. MPC = 0.5

a (autonomous consumption) = ₹200 crores

I = ₹160 crores

$Y = C + I$

$Y = a + bY + I$

$Y = 200 + 0.5(Y) + 160$

$Y = 360 + 0.5\,Y$

$Y - 0.5\,Y = 360$

$0.5\,Y = 360$

$Y/2 = 360$

Y = ₹ 720 Crores

Q.3. Calculate the value of investment expenditure from the following data if an economy is in equilibrium:

Marginal propensity to consume = 0.8

Autonomous consumption=100

Level of income=2000 [OEB]

Ans. $Y = C + I$

$Y = a + bY + I$

$2000 = 100 + 0.8(2000) + I$

$2000 - 100 = 8/10(2000) + I$

$1900 = 1600 + I$

$1900 - 1600 = I$

I = ₹300 crores

This question is for practice and its solution is available at the end of the chapter

Q.4. Give difference between average propensity to consume and the marginal propensity to consume using a numerical example. [OEB]

Ans. Average propensity to consume can be defined as the ratio of consumption expenditure and income. Marginal propensity to consume can be defined as the ratio of change in consumption expenditure and change in income.

APC = C/Y

Income	Consumption	APC
100	120	1.2
200	160	0.8

MPC = ΔC/ΔY

Income	Consumption	MPC
100	120	-
200	160	40/100 = 0.4

6 Marks Questions

Q.1. What is the difference between MPC and MPS? How are they related?

Concept Applied	Marginal propensity to consume and Marginal propensity to save

Ans. Marginal Propensity to Consume is the ratio of a Change in Consumption to a Change in Income. It is calculated as under :

$$MPC = \frac{\text{Change in Consumption}}{\text{Change in Income}} = \frac{\Delta C}{\Delta Y}$$

Marginal Propensity to Save is the ratio of a Change in Savings to a Change in Income. It is calculated as under :

$$MPS = \frac{\text{Change in Savings}}{\text{Change in Income}} = \frac{\Delta S}{\Delta Y}$$

The relationship between the two is as under:

$$MPC + MPS = \frac{\Delta C}{\Delta Y} + \frac{\Delta S}{\Delta Y} = \frac{\Delta C + \Delta S}{\Delta Y} = \frac{\Delta Y}{\Delta Y} = 1$$

$$\Rightarrow MPC + MPS = 1$$

Q.2. Distinguish between Marginal Propensity to Consume and Average Propensity to Consume. Give a numerical example. [OEB]

Ans.

Average Propensity to Consume	Marginal Propensity to Consume
Average propensity to consume refers to the ratio of Consumption Expenditure to the corresponding level of income.	Marginal propensity to consume refers to the ratio of change in Consumption Expenditure (ΔC) to change in total income (ΔY).
$APC = \frac{\text{Consumption}}{\text{Income}}$	$MPC = \frac{\text{Change in consumption}}{\text{Change in income}}$
It can be greater than 1, equal to 1 or less than 1.	Its value lies between 0 and 1.
It can never be zero.	Its value can be zero.
Ex. Supposing at a given level of income of 300 crore consumption is 250 crore then APC will be 250/300 which is equal to 0.83.	Ex. Suppose income increases by 100 crore and consumption increases by 50 crore then MPC = 50/100 which is equal to 0.5.

Q.3. Differentiate between Aggregate Demand and Aggregate Supply. [OEB]

Ans. Aggregate Demand : It refers to the total value of final goods and services which all the sectors of an economy are planning to buy at a given level of income during a period of an accounting year.

Aggregate Supply : It refers to the money value of goods and services that all the producers are willing to supply in an economy in a given time period.

Topic-2 Short-run Equilibrium and Concept of Multiplier

Concepts Covered: Short Run Equilibrium, Change in Equilibrium, Multiplier, Relationship between Multiplier and MPC, Paradox of Thrift

Revision Notes

- **Short Run :** According to **J. M. Keynes**, "A period of time during which level of output is determined exclusively by the level of employment in the economy, is termed as short run."

- **Classical Concept of Equilibrium :** It is based on **J.B. Say's** Law, " Supply creates its own demand". Main contribution of classical concept of equilibrium are as follows :

(i) Every economy works at full employment level.
(ii) Economy has a capacity of self-adjustment. Whatever is produced in an economy is automatically consumed, i.e., Supply Creates its own demand.
(iii) Over production in an economy is impossible, because every additional production generates additional income and as a result total income equals total saving.
(iv) General unemployment in the economy is not possible because no possibility of over production arises in the economy.
(v) Wage flexibility helps in elimination of unemployment.
(vi) Interest rate flexibility maintains the equality between saving and investment.

- **Short Run Equilibrium, i.e., Keynesian Approach :**

(a) AD = AS Approach
Employment is determined at a point where AD = AS.
If AD > AS, firm will employ more factors of production and it will again attain AD = AS.
If AD < AS, firm will cut employment and it will bring again AD = AS.

(b) S = I Approach
According to Keynes income-employment equilibrium is determined at a point where S = I.
If S > I then equilibrium income will have a tendency to reduce.
If S < I then equilibrium income will have a tendency to increase.

- **Ex-Post (S) = Ex-Post (I) :** At all levels of income and they are necessarily be equal.
- **Ex-Ante (S) = Ex-Ante (I) :** Only at equilibrium level of income.
- **Change in Equilibrium :** Equilibrium position described above may be of full employment or may not be of full employment. It only determines the level of income. Therefore, for full employment we have to twist AD or AS. But AS depends on technological factors therefore if AD increases, it will raise the level of employment.
- **Multiplier :** It establishes relation between investment and income. It measures the change in income due to change in investment.

$$K = \frac{\Delta Y}{\Delta I} = \frac{\text{Change in Income}}{\text{Change in Investment}}$$

- **Relationship between Multiplier and Marginal Propensity to Consume (MPC) :** The size of multiplier is determined by the Marginal Propensity to Consume. There is a direct relation between MPC and K. Higher the MPC, higher is value of K and *vice-versa*.

$$K = \frac{1}{1 - MPC}$$

and, $K = \frac{1}{MPS}$

if MPC = Zero, K = 1
if MPC = 1 K = ∞

Value of K lies between 1 and infinity.

- **Paradox of Thrift :** According to this Paradox, if all the consumers in an economy increase the proportion of their saving, that is, if MPS increases, still the overall savings in the economy will either remain same or decrease but will never increase.

Example:

1. If the value of MPC is 075, find the value of multiplier.

Ans. Given, MPC = 0.75

$$K = \frac{1}{1 - MPC} = \frac{1}{1 - 0.75} = \frac{1}{0.25} = 4$$

2. In an economy, how much additional investment should be generated in the private sector to raise national income by ₹150 Cr. while MPC is 0·5?

Ans. Given,

ΔY = ₹150 crore
MPC = 0.5

$$K = \frac{1}{1 - MPC} = \frac{1}{1 - 0.5} = \frac{1}{0.5} = 2$$

$$K = \frac{\Delta Y}{\Delta I}$$

$$2 = \frac{150}{\Delta I}$$

$$\Delta I = \frac{150}{2} = ₹75 \text{ crore.}$$

Key Terms

- **Ex-Ante Saving :** It is the planned or desired or intended saving during a particular period.
- **Ex-Ante Investment :** It is the planned or desired or intended investment during a particular period.
- **Ex-Post Saving :** It is realised saving. It is equal to Ex-Ante saving + Unplanned saving.
- **Ex-Post Investment :** It is realised investment. It is equal to Ex-Ante investment + Unplanned investment.

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. In an economy, how much additional investment should be generated in the private sector to raise national income by ₹ 450 Cr. while MPC is 0·6? [ISC Specimen Paper 2023-24]

Concept Applied Investment Multiplier

Ans. Given,

ΔY = ₹450 crore

MPC = 0.6

$$K = \frac{1}{1-MPC} = \frac{1}{1-0.6} = \frac{1}{0.4} = 2.5$$

$$K = \frac{\Delta Y}{\Delta I}$$

$$2.5 = \frac{450}{\Delta I}$$

$$\Delta I = \frac{450}{2.5} = ₹180 \text{ crore.}$$

Q.2. How does an increase in national income affect the income-elastic investment in the economy?
[ISC Specimen Paper 2023-24]

Ans. The investment expenditure will increase in the case of the income-elastic investment when there is an increase in the national income. This will lead to increase in the aggregate demand and so may lead to the situation of inflation in the economy.

Q.3. Full employment does not necessarily imply zero unemployment. Justify.
[ISC Specimen Paper 2023-24]

Ans. Full employment is the situation when all the able-bodied people who are willing to work, are employed at the prevailing income. Zero unemployment on the other hand means that no person who is the part of the work force is unemployed. Thus, there are instances that the person may be in training or enjoying leisure time, so there will be no zero employment but there is full employment as the person is employed.

Q.4. Give the meaning of gross investment? [OEB]

Ans. Gross investment can be defined as the investment made by all the producers in the economy and it includes depreciation only.

Q.5. Calculate the value of multiplier when the value of MPC is 1. [OEB]

Ans. Multiplier (K) = 1/(1 – MPC)
K = 1 / (1 – 1)
K = 1/ 0
K = ∞

Q.6. State giving reason whether the following statement is true or false: [OEB]
Consumption changes in the same direction as income. It is,

Ans. True.

Q.7. "When planned investment is less than planned saving then there is accumulation of unplanned inventories." [OEB]

Ans. True
Reason: When planned investment is less than planned saving, then planned inventory would fall below the desired level. To bring back the inventory to the desired level, producers expand the output and hence more income is generated.

Q.8. "When planned investment is more than planned saving, then there is accumulation of unplanned inventories." [OEB]

Ans. True
Reason: Unplanned inventories accumulate when planned investment is less than planned saving. Planned savings is the intended or planned savings by the households at the beginning of a particular period of time.

2 Marks Questions

Q.1. If the initial increase in autonomous investment is ₹100 cr and MPC = 0.8, calculate the increase in income. [ISC Semester-II, 2022]

Ans.

$$K = \frac{1}{1-MPC}$$

$$= \frac{1}{1-0.8} = \frac{1}{0.2} = 5$$

$$K = \frac{\Delta Y}{\Delta I}$$

$$5 = \frac{\Delta Y}{100}$$

$$\Delta Y = 5 \times 100$$

$$\Delta Y = ₹500 \text{ Crore.}$$

Q.2. What is meant by investment multiplier?
[ISC 2019]

Ans. Investment multiplier: If there is increase in autonomous investment in an economy, then the real national income will increase by a multiple of that initial increase in autonomous investment. / defined as the multiple amount by which income increases as a result of increase in investment expenditure.

$$K = \frac{\Delta Y}{\Delta I}$$

$$K = \frac{1}{1-MPC} \text{ OR } \frac{1}{MPS}$$

(meaning and formula)
[ISC Marking Scheme, 2019]

Examiner's Comment

- Many candidates were unable to explain the investment multiplier effect according to which income increases in investment. Several candidates instead of mentioning the multiple amounts, wrote the formula for investment multiplier in words.

Answering Tip

- Emphasize that the definition of investment multiplier should be accompanied with its formula.

Q.3. If the value of MPC is 0·9, find the value of multiplier. [ISC 2016]

Concept Applied Multiplier

Ans. Given, MPC = 0.9

$$K = \frac{1}{1-MPC} = \frac{1}{1-0.9} = \frac{1}{0.1} = 10$$

Examiner's Comment

- A number of candidates lost marks as they did not know the formula of multiplier.

Answering Tip

- Learn the concept of multiplier properly K = 1/1-MPS or K = 1/1-MPC

Q.4. An economy is in equilibrium. Calculate the Investment Expenditure from the following :
National Income = 16,000
Marginal Propensity to Save = 0.25
Autonomous Consumption = 2,000 [OEB]
Concept Applied: Short Run Equilibrium

Ans. Given, Y = 16,000
MPS = 0.25
or MPC = 1 – MPS
= 1 – 0.25 = 0.75
C = 2,000
We know, Y = C + b(Y) + I
or 16,000 = 2,000 + 0.75 (16,000) + I
or 16,000 = 2,000 + 12,000 + I
or I = 16,000 – 14,000
or I = 2,000

Q.5. What will be the value of the multiplier if the marginal propensity to consume and marginal propensity to save are equal? [OEB]

3 Marks Questions

Q.1. Briefly explain the mechanism of the investment multiplier with the help of a schedule. Take initial increase in autonomous investment as ₹ 1000 crores and MPC as 0·8. [ISC 2020] AI

OR

Discuss the mechanism of investment multiplier with the help of a numerical example. [ISC 2018]

Concept Applied Investment Multiplier

Ans. Investment multiplier is the multiple amounts by which income increases as a result of increase in investment expenditure.

$$K = \frac{\Delta Y}{\Delta I} = \frac{1}{1-MPC} = \frac{1}{MPS}$$

Rounds of spending	Increase in autonomous investment (ΔI)	Increase in consumption (ΔC)	Increase in income (ΔY)
1	1000	-	1000
2	-	0.8 × 1000 = 800	800
3	-	0.8 × 800 = 640	640
4	-	0.8 × 640 = 512	512
		(ΔC) = 4000	(ΔY) 5000

Given: autonomous investment increases by ₹ 1000 crores and MPC = 0.8

- As investment doesn't affect consumption in the first round there is no change in C, but Y increases by 1000 (as Y = C+I).
- From round 2, C starts increasing by MPC x ΔY.
- The process continues till ΔC and ΔY are too small to continue.
- Therefore, due to increase in autonomous investment by ₹ 1000 income increases by ₹ 5000.

(This final amount of ΔY could also be summarized by appropriate mathematical formula.)

[ISC Marking Scheme, 2020]

Examiner's Comment

- Most candidates concentrated only on the formula while ignoring the definition of the multiplier. Many candidates calculated the total increase in the income by only using the formula. They did not write the mechanism and did not mention the maximum amount of income that can be created in the process.

This question is for practice and its solution is available at the end of the chapter

Answering Tip

- The students need to ensure that they write the definition of the multiplier along with its formula. They should arrive at the change in income and consumption with the help of a table. Students should write the process of investment multiplier along with the formula. The students should attempt the question on mechanism of the investment multiplier with a schedule and if values are given then only those values should be used.

Q.2. Find the value of additional investment made by the government, when MPC = 0·5 and increase in income (ΔY) = ₹1000. [ISC 2015]

Ans. Given, MPC = 0.5

ΔY = ₹1000

$$K = \frac{1}{1-MPC} = \frac{1}{1-0.5} = \frac{1}{0.5} = 2$$

$$K = \frac{\Delta Y}{\Delta I}$$

$$2 = \frac{1000}{\Delta I}$$

$$\Delta I = \frac{1000}{2} = ₹500$$

Thus, an additional investment of ₹500 will be made.

Examiner's Comment

- Many candidates attempted the numerical correctly. In some cases the formula used was incorrect. A few candidates solved only till the multiplier and left the value of additional investment.

Answering Tip

- Analyse the investment multiplier mechanism clearly.

Q.3. Give the meaning of multiplier. State the relation between MPC and multiplier. Calculate the value of MPC if the value of multiplier is 2. [OEB]

Ans. Multiplier can be defined as the ratio of change in income to change in investment.

There is direct relation between the value of MPC and multiplier.

Multiplier = 1/ (1 - MPC)

If MPC = 0.6

Then, Multiplier = 1/ (1 – 0.6)

= 1/ 0.4

= 2.5

If MPC = 0.8

Then, multiplier = 1/(1-0.8)

= 1/0.2

= 5

Multiplier = 1/ (1 - MPC)

2 = 1/ (1-MPC)

2(1 - MPC) = 1

2 – 2MPC = 1

2 – 1 = 2 MPC

1 = 2 MPC

MPC = ½ = 0.5

Q.4. In an economy, there is increase in investment by ₹1000 and as a result of it there is increase in income by ₹5500. If marginal propensity to save is 50%, how much is there increase in income? [OEB]

Ans. ΔI = ₹ 1000

ΔY = ₹ 5500

MPS = 0.50

K = 1/ MPS

K = 1/0.50

K = 2

K = ΔY/ΔI

K = ΔY/1000

2 = ΔY/1000

ΔY = ₹ 2000

Q.5. State the range of value of investment multiplier? State the relation of investment multiplier with marginal propensity to consume (MPC) and with marginal propensity to save (MPS). [OEB]

Ans. Value of Investment Multiplier:

(i) Minimum value of multiplier is 1 because minimum value of MPC can be zero.

(ii) Maximum value of multiplier may be–(infinite) because maximum value of MPC can be 1.

Relationship of K (investment multiplier) with MPC and MPS:

(i) There is direct relationship between k and MPC. If MPC is high, k will also be high but if MPC is low, k will also be low. Thus, k varies directly with value of MPC. In short, higher the value of MPC, higher will be the value of multiplier and vice versa.

(ii) There is inverse relationship between k and MPS. If MPS is high, k will be low but if MPS is low, k will be high.

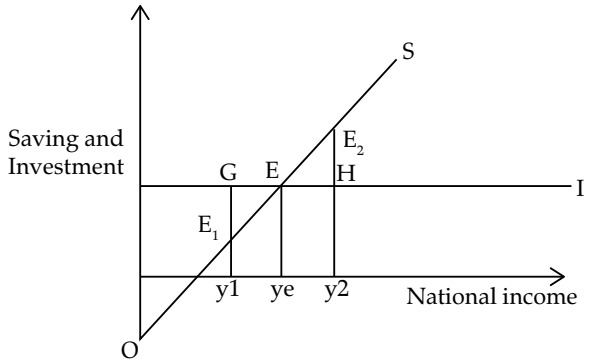
4 Marks Questions

Q.1. Determine the output and equilibrium level of income with the help of Saving and Investment approach. [ISC Specimen Paper 2023-24]

OR

Explain how equilibrium level of income can be determined with the help of saving and investment approach. [ISC 2019]

Ans. According to Keynes income-employment equilibrium is determined at a point where Planned savings is equal to planned Investment (S = I).



In the given diagram, the investment curve is parallel to X-axis as we assume that there is autonomous investment which remains same. The point where S = I is the point where National Income Y is determined, that is the point of intersection E.

(i) If S > I the consumers are consuming less, leading the producers to reduce the production. This leads to the decrease in the inventories. The equilibrium income will have a tendency to reduce till the S = I.

(ii) If S < I, the consumers are consuming more. This leads the producers to be short of the inventories leading them to increase the production. This in turn increases the inventories and then equilibrium income will have a tendency to increase until the savings equals investments.

Examiner's Comment

- Many candidates drew the diagrams for AD-AS approach and S-I approach in upper panels and lower panels but did not show the correspondence of the equilibrium points of these two diagrams. Some candidates explained the answer in terms of AD-AS approach but drew the diagram for S = I approach. A few candidates drew the Saving curve passing through the origin. The diagram was not properly labelled in a few answer scripts.

Answering Tips

- Learn the Saving and Investment approach for equilibrium level by properly drawing the Saving curve which starts from below the x-axis, considering only autonomous investment.
- Learn this approach when S<I, S = I and S>I and its consequent effect on the income and employment. For better understanding by the students, this approach should be explained with the help of AD and AS approach.
- The students need to understand that a definition should be followed by its mathematical formula.

Q.2. Explain the concept of Investment Multiplier using a diagram. A I **[ISC 2023]**

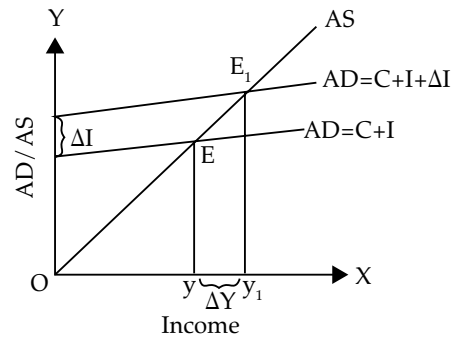
Concept Applied | Investment Multiplier

Ans. (i) Investment multiplier is the ratio of an increase of income to given increase in investment. Investment multiplier refers to the increase in the aggregate income of the economy as a result of an increase in the investments done by the government in the form of new projects.

The formula of investment multiplier can be expressed as:

$$K = \frac{\Delta Y}{\Delta I}$$

Graphically investment multiplier can be represented as follows:



Q.3. Calculate change in final income, if Marginal Propensity to Consume (MPC) is 0.8 and change in initial investment is ₹ 5,000 crore. [OEB]

Ans. Investment Multiplier (K)

$$= \frac{1}{1-MPC} = \frac{1}{1-0.8} = \frac{1}{0.2} = 5$$

Also, $K = \frac{\Delta Y}{\Delta I}$

$$5 = \frac{\Delta Y}{5000}$$

ΔY = 25,000 crore

Change in final income = ₹25,000 crore

Q.4. Estimate the change in initial investment if Marginal Propensity to Save (MPS) is 0·25 and change in final income is ₹95,000 crore. [OEB]

Ans. Investment Multiplier (K) = $\frac{1}{MPS}$

$$K = \frac{1}{0.25} = 4$$

Also, $K = \frac{\Delta Y}{\Delta I}$

$$4 = \frac{95,000}{\Delta I}$$

ΔI = ₹23,500 crore

Change in final income = ₹23,500 crore

Q5. Estimate the change in final income if Marginal Propensity to Consume (MPC) is 0·9 and change in initial investment is ₹32,000 crore. [OEB]

This question is for practice and its solution is available at the end of the chapter

6 Marks Questions

Q.1. Explain how equilibrium level of income can be determined with the help of aggregate demand curve and aggregate supply curve. [ISC 2017]

Ans. According to the Keynesian Theory, equilibrium condition is generally stated in terms of aggregate demand (AD) and aggregate supply (AS). An economy is in equilibrium when aggregate demand for goods and services is equal to aggregate supply during a period of time.

So, equilibrium is achieved when:

AD = AS … **(1)**

We know, AD is the sum total of Consumption (C) and Investment (I):

AD = C + I … (2)

Also, AS is the sum total of consumption (C) and saving (S):

AS = C + S … (3)

Substituting (2) and (3) in (1), we get:

C + S = C + I

Or, S = I

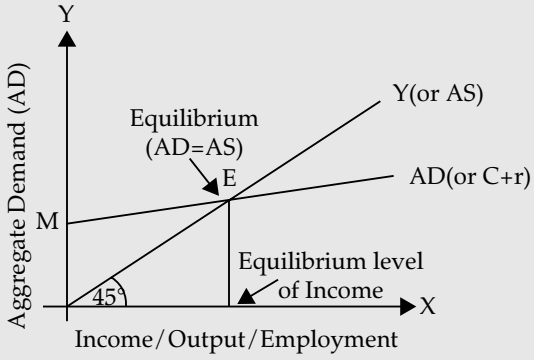
It means, according to Keynes, there are Two Approaches for determining the equilibrium level of income and employment in the economy:

Aggregate Demand-Aggregate Supply Approach (AD-AS Approach):

According to the Keynesian theory, the equilibrium level of income in an economy is determined when aggregate demand, represented by C + I curve is equal to the total output (Aggregate Supply or AS).

Components of Aggregate Supply:

Aggregate supply is the total output of goods and services of the national income. It is depicted by a 45° line. Since the income received is either consumed or saved, the AS curve is represented by the (C + S) curve.



In the figure, the AD or (C +1) curve shows the desired level of expenditure by consumers and firms corresponding to each level of output. The economy is in equilibrium at point 'E' where (C + I) curve intersects the 45° line.

(i) 'E' is the equilibrium point because at this point, the level of desired spending on consumption and investment exactly equals the level of total output.

(ii) OY is the equilibrium level of output corresponding to point E.

(iii) It is a situation of 'Effective Demand'. Effective demand refers to that level of AD which becomes 'effective' because it is equal to AS.

At y_1, aggregate demand is greater than aggregate supply so inventory will be used up, income will increase to y.

At y_2, aggregate supply will be greater than aggregate demand and income will decrease as inventory will increase.

[ISC Marking Scheme, 2017]

Q.2. Answer the following questions based on the data given below :

(i) Planned Investments = ₹ 1000 crores.

(ii) C = 500 + 0.50 Y

(a) Determine the equilibrium level of income.

(b) Calculate the value of Savings at equilibrium level of National Income.

(c) Calculate the value of Investment Multiplier.

Ans. (a) At equilibrium

AD = Y = C+I

thus, Y = (500 + 0.5Y) + 1000

Y – 0.5 Y = 1500

$$Y = \frac{1500}{0.5} = ₹3000 \text{ crores.}$$

So, equilibrium level of income = ₹ 300 crores.

(b) Consumption at equilibrium level of income

S = (–) c + (1–b)Y

S = (–)500 +(0.5)(3000)

Savings = ₹1000 crores (any other alternative correct method to be awarded marks) **[½]**

(c) Investment Multiplier = $\frac{1}{1-MPC}$

$= \frac{1}{1-0.5}$

= 2

Topic-3 Concepts of Excess Demand and Deficient Demand

Concepts covered: Full Employment, Under Employment, Involuntary Employment, Deficient Demand, Deflationary Gap, Inflationary Gap

Revision Notes

- **Full Employment :** It refers to a situation, where all those workers who are able to work and willing to work get employment at prevailing wage rate.
- **Under Employment :** It refers to a situation, where those workers who are able to work and willing to work do not get employment at prevailing wage rate.
- **In an economy :**
- Income Equilibrium Level = Output Equilibrium Level = Employment Equilibrium Level
  (Y) (O) (N)
- Voluntary unemployment occur when some people are not willing to work at all or are not willing to work at the existing wage rate.
- **Involuntary Unemployment :** It is that part of the labour force which is willing and able to work at the prevailing wage rate but is out of work. **Significance :** It helps to estimate Aggregate Demand.
- **Deficient Demand :** It refers to a situation in which Aggregate Demand remains short of required Aggregate Supply corresponding to full employment.
  Deficient Demand = AD < AS (at full employment level).
- **Deflationary Gap or Measurement of Deficient Demand :** Deflationary Gap is the measurement of deficient demand and is equal to the difference between Aggregate Demand at full employment (AD_F) and actual Aggregate Demand.

$$\text{Deflationary Gap} = AD_F - AD$$

- **Reasons for Deficient Demand :** (i) Reduction in Supply of Currency, (ii) Increase in Bank Rate, (iii) Increase in Taxes, (iv) Reduction in Public Expenditure, (v) Increase in Propensity to Save, (vi) Decline in Export Demand.
- **Effects of Deficient Demand :** (i) Fall in production level, (ii) Fall in price level (iii) Increase in unemployment.
- **Measures to correct Deficient Demand:**

1. **Fiscal Policy :** (a) Increase in Expenditure, (b) Reduction in Tax burden, (c) Reduction in Public Debt, (d) Deficit Budget
2. **Monetary Policy :** (a) Reduction in Bank Rate, (b) Purchase of Securities in Open Market, (c) Reduction in Cash Reserve Ratio, (d) Reduction in Liquidity Ratio, (e) Reduction in Margin Requirement of Money, (f) Abolition of Credit Rationing, (g) Direct Action
3. **Others :** (a) Less imports, more exports, (b) Stable Wage Policy, (c) Government purchases at higher support price.

- **Excess Demand :** When current Aggregate Demand becomes higher than the amount of Aggregate Demand required for full employment.

AD > AS (at full employment level)

- **Inflationary Gap or Measurement of Excess Demand:** Inflationary gap is the measurement of excess demand and is equal to the difference between Aggregate Demand beyond full employment (AD) and Aggregate Demand at full employment (AD_F).

$$\text{Inflationary Gap} = AD - AD_F$$

- **Reasons for Excess Demand :** (i) Increase in public expenditure, (ii) Reduction in taxes, (iii) Deficit financing, (iv) Extension of credit facilities, (v) Increase in investment demand, (vi) Increase in propensity to consume (vii) Increase in export demand.
- **Effects of Excess Demand :** (i) Effects on employment, (ii) Effects on production, and (iii) Effects on prices.
- **Measures to correct Excess Demand:**

1. **Fiscal Policy :** (a) Reduction in Public Expenditure, (b) Increase in tax burden, (c) Increase in Public Debt, (d) Surplus Budget
2. **Monetary Policy :** (a) Rise in Bank rate, (b) Sale of Securities in Open Market, (c) Rise in Cash Reserve Ratio, (d) Increase in Liquidity Ratio, (e) Increase in Margin Requirement of Money, (f) Credit Rationing, (g) Direct Action
3. **Others :** (a) More import, less exports, (b) Linking of wages with productivity, (c) Increase in production.

Key Terms

- **Fiscal Policy :** Fiscal policy refers to revenue and expenditure policy of the government, to correct the situation of excess and deficient demand in the economy.
- **Monetary Policy :** It is that policy of the government which corrects the situation of excess and deficient demand by regulating interest rate and availability of credit in the economy.

- **Varies measures of monetary policy are :**

(i) **Quantitative :** Bank Rate, Open Market Operations, Minimum Reserve Ratio and Liquidity Ratio.

(ii) **Qualitative :** Margin requirement of loans, Credit rationing, Direct action and Moral suasion.

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. How is deflationary gap generated in the economy? [ISC Specimen Paper 2023-24]

Ans. When the aggregate demand falls short of the aggregate supply in the economy, it causes the situation of deflationary gap.

Q.2. Give the meaning of deficient demand. [ISC 2023]

Ans. A situation when the Aggregate Demand is less than the Aggregate Supply in an economy, corresponding to full employment in the economy, is known as deficient demand.

Q.3. Why is effective demand also known as ex-post demand? [ISC 2023]

Ans. An effective demand is also known as ex-post demand. Effective demand means the amount of goods that consumers are actually buying. This aspect of demand is considered as both effective and ex-post demand.

Q.4. Give the meaning of excess demand? [OEB]

Ans. Excess demand can be defined as when aggregate demand is more than aggregate supply at the full employment level of income.

Q.5. Give the meaning of deficient demand? [OEB]

Q.6. State giving reason whether the following statement is true or false: [OEB]
"Deflationary gap can be defined as when aggregate demand is greater than aggregate supply at full employment level."

Ans. False
Reason: Deflationary gap can be defined as the gap in which aggregate demand is less than aggregate supply at full employment level of income.

Q.7. Full employment mean zero unemployment. [OEB]

Ans. False.
Reason: Even in the situation of full employment, natural rate of unemployment always exits in the economy.

Q.8. Voluntary unemployment occurs in the economy under deflationary gap. [OEB]

Ans. False.
Reason: In the situation of deflationary gap, involuntary unemployment exists.

Q.9. Under employment implies voluntary unemployment. [OEB]

Ans. False.
Reason: Under employment implies involuntary unemployment where workers are ready to work at prevailing wage rate but not getting the work.

Q.10. Excess demand raises the market value of output. [OEB]

Ans. True.
Reason: Because higher demand than supply leads to decrease in general price level.

2 Marks Questions

Q.1 State the difference between excess demand and deficient demand? [OEB]

Ans. Excess demand can be defined as when aggregate demand is greater than aggregate supply at the full employment level of income. Deficient demand can be defined as aggregate demand is less than aggregate supply at the full employment level of income.

Q.2. How can Repo rate be used to curb the inflationary gap? [OEB]

Ans. Repo rate is the rate at which the Central Bank lends money to the Commercial Banks. To correct the situation of Inflationary Gap, Repo Rate is increased. As a follow-up action, Commercial Banks raise the market rate of interest (the rate at which the Commercial Banks lends money to the consumers and the investors). This reduces demand for credit. Consequently, consumption expenditure and investment expenditure get reduced. Implying a reduction in Aggregate Demand, as required to correct Inflationary Gap.

Q.3. How does decreasing the taxes help in correcting the situation of deficient demand? [OEB]

Ans. To curb the deflationary gap the government may decrease the taxes. This may increase the purchasing power in the hands of the public which in turn may increase the Aggregate Demand in the economy to bring it equal to the Aggregate Supply.

Q.4. "The RBI can use open market operations to reduce the inflationary gap." Justify. [OEB]

Ans. Sale of Government Securities in the open market by the Central Bank will adversely affect the ability of the Commercial Banks to create credit in the economy. As a result, Aggregate Demand in the economy may fall and correct the inflationary gap in the economy.

This question is for practice and its solution is available at the end of the chapter

Q.5. Government borrowings can lead to the correction of excess demand. How? [OEB]

Ans. The government in order to curb the inflationary gap situation can borrow more internal debts. This will decrease the money supply in the economy, decreasing the purchasing power of the people. As the consumption expenditure will decrease, it will lead to a decrease in the aggregate demand to bring it equal to the Aggregate Supply.

3 Marks Questions

Q.1. Discuss any two fiscal measures to correct a situation of deficient demand in an economy. [ISC 2019]

Ans. The fiscal measures that can be used to correct the situation of deficient demand in an economy are:

(i) Decrease in Taxes: To curb the deflationary gap the government may decrease the taxes. This may increase the purchasing power in the hands of the public which in turn may increase the Aggregate Demand in the economy to bring it equal to the Aggregate Supply.

(ii) Increase in Government Expenditure: To curb the deflationary gap the government may increase its non-developmental expenditure. This may increase the purchasing power in the hands of the people which in turn will increase the Aggregate Demand in the economy to bring it equal to the Aggregate Supply.

(iii) Decrease in Public Debt: The government in order to curb the deflationary gap situation can repay the internal borrowings. This will increase the money supply in the economy, increasing the purchasing power of the people. As the consumption expenditure will increase, it will lead to an increase in the aggregate demand to bring it equal to the Aggregate Supply.

(iv) Increase in Deficit Financing: The government can take more loans to finance its government expenditure. This will lead to increase in the consumption and investment expenditure of government, thus increasing the aggregate demand. Ultimately the aggregate demand will equal the aggregate supply curbing the situation of deflationary gap. **[Any Two]**

Examiner's Comment

- Many candidates mentioned the correct fiscal measures to correct a situation of deficient demand in an economy but were unable to explain how these measures, especially measures such as taxation, public expenditure and public borrowing, generated aggregate demand.

Answering Tips

- Learn all fiscal measures to control different economic conditions to the students.
- The students need to know that use of disposable income is appropriate in case of taxation but not for public expenditure.
- With proper logic, analyse the use of public borrowing to generate demand.

Q.2. Explain how fiscal policy measures can be used to reduce excess demand in an economy. [ISC 2017]

Ans. The fiscal policy measures that can be used to reduce the excess demand in an economy are:

(i) Increase in Taxes: To curb the inflationary gap the government may increase the taxes. This may reduce the purchasing power in the hands of the public which in turn may reduce the Aggregate Demand in the economy to bring it equal to the Aggregate Supply.

(ii) Reduction in Government Expenditure: To curb the inflationary gap the government may reduce its non-developmental expenditure. This may reduce the purchasing power in the hands of the people which in turn will reduce the Aggregate Demand in the economy to bring it equal to the Aggregate Supply.

(iii) Increase in Public Debt: The government in order to curb the inflationary gap situation can borrow more internal debts. This will decrease the money supply in the economy, decreasing the purchasing power of the people. As the consumption expenditure will decrease, it will lead to a decrease in the aggregate demand to bring it equal to the Aggregate Supply.

(iv) Reduction in Deficit Financing: The government reduce financing its government expenditure. This will lead to decrease in the consumption and investment expenditure of government, thus decreasing the aggregate demand. Ultimately the aggregate demand will equal the aggregate supply curbing the situation of inflationary gap. **[Any Two]**

Examiner's Comment

- Many candidates answered this part correctly. However, under fiscal policy, several candidates wrote monetary policy. A few candidates mentioned the role of public debt and deficit financing.

Answering Tip

- The students should learn how increase in taxes, reduction in public expenditure, increase in public borrowing and reduction in deficit financing would reduce excess demand.

Q.3. Excess demand is caused due to a number of reasons. Explain any three of them. [OEB]

Concept Applied	Reasons for Excess Demand

Ans. The reason for excess demand are:

(i) Increase in public expenditure: When the government increases the public expenditure, it

increases the aggregate demand in the economy by raising the income level which raises the consumption expenditure.

(ii) **Reduction in taxes:** When the government reduces the taxes in the economy, the real income with the consumers increases. This leads them to increase the consumption expenditure leading to an increase in the aggregate demand above the equilibrium level.

(iii) **Deficit financing:** When the government borrows money to finance its deficit budget, it means that the government expenditure is in excess. As public expenditure is a part of Aggregate demand, meaning that aggregate demand in the economy rises.

(iv) **Increase in investment demand:** Increase in Investment expenditure leads to an increase in aggregate demand of the economy as investment is a part of aggregate demand.

(v) **Increase in propensity to consume:** Propensity to consume directly effects the consumption expenditure in the economy.

(vi) **Increase in export demand:** When the export in the economy rises, it leads to an increase in the income of the domestic firms. This leads to an increase in the consumption expenditure. **[Any Three]**

Q.4. "Deficient demand is not at all good for the economy." What are the causes of such a situation in an economy? [OEB]

Ans. The reasons for deficient demand are:

(i) **Decrease in public expenditure:** When the government decreases the public expenditure, it decreases the aggregate demand in the economy by decreasing the income level which decreases the consumption expenditure.

(ii) **Increase in taxes:** When the government increases the taxes in the economy, the real income with the consumers decreases. This leads them to decrease the consumption expenditure leading to a decrease in the aggregate demand.

(iii) **Decrease in investment demand:** Decrease in Investment expenditure leads to a decrease in aggregate demand of the economy as investment is a part of aggregate demand.

(iv) **Decrease in Money Supply:** If the money supply decreases in the economy, it leads to the fall in the aggregate demand.

(v) **Decrease in propensity to consume:** Propensity to consume directly effects the consumption expenditure in the economy.

(vi) **Decrease in export demand:** When the export in the economy falls, it leads to a decrease in the income of the domestic firms. This leads to a decrease in the consumption expenditure. **[Any Three]**

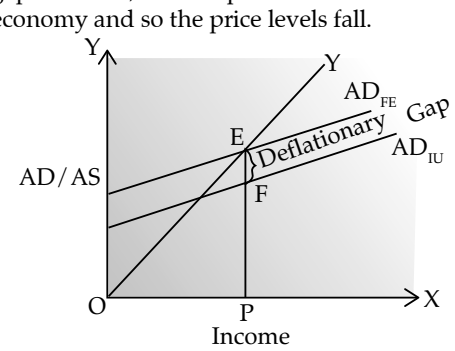
Q.5. Define margin requirement. State the measure to control deflation in economy by the central bank? [OEB]

6 Marks Questions

Q.1. Explain the concept of deflationary gap with the help of a diagram. Discuss any two monetary measures to reduce the gap. [ISC 2020]

Ans. Deficient demand refers to a situation when aggregate demand is less than aggregate supply corresponding to full employment.

When aggregate demand is lower than aggregate supply at full employment, the gap is called deflationary Gap. That is, it occurs when Aggregate Demand is smaller than Aggregate Supply (AD<AS) at the full employment level. In the deflationary gap situation, there is prevalence of deflation in the economy and so the price levels fall.



From the figure we can see that the AD_{FE} is the aggregate demand curve at the full employment level and AD_{IU} is the aggregate demand curve below full employment level. The point E is the equilibrium point where AD = AS.

But the deficient demand (current) of ADIU, aggregate demand FP is less than the aggregate supply in the economy. This difference of actual aggregate demand and supply i.e., EF is the deflationary gap.

Deflationary Gap = Deficient Demand = ADFE – ADIU = EF

The monetary measures to overcome the situation of deflationary gap are:

(i) **Decrease in Cash Reserve Ratio:** A decrease in Cash Reserve Ratio (CRR) may increase the credit creation capacity of the commercial banks in the economy. This may lead to a rise in the borrowings from banks causing a rise in Aggregate Demand in the economy and helps to correct the deflationary gap in the economy.

(ii) **Decrease in Repo Rate:** Repo rate is the rate at which the Central Bank lends money to the Commercial Banks. To correct the situation of Deflationary Gap, Repo Rate is decreased. As a follow-up action, Commercial Banks decreases the market rate of interest (the rate at which the Commercial Banks lends money to the consumers and the investors). This increases demand for credit. Consequently, consumption expenditure and investment expenditure get increased. Implying a rise in Aggregate Demand, as required to correct Deflationary Gap.

This question is for practice and its solution is available at the end of the chapter

(iii) Decrease in Reverse Repo Rate: Reverse Repo Rate is the rate of interest at which commercial banks can park their surplus funds with the central bank, for a relatively shorter period of time. To deal with the situation of deficit demand this rate may be decreased by the central bank. It may discourage the commercial bank to park their surplus funds with Central Bank. As a result, the availability of creating credit with the commercial bank will be increased. Consequently, consumption expenditure and investment expenditure may get increased, implying reduction in Aggregate Demand.

(iv) Open Market Operations: Purchase of Government Securities in the open market by the Central Bank will positively affect the ability of the Commercial Banks to create credit in the economy. As a result Aggregate Demand in the economy may rise and correct the deflationary gap in the economy.

(v) Margin Requirement: Margin requirement is the difference between amount of loan offered and the market value of security offered against the loan. Lower the margin requirement more will be the demand for loan. In this situation of deficit demand, there is a need to increase loan/credit which can be done by lowering the margin requirements. **[Any Two]**

Examiner's Comments

- **The common errors made by the candidates were:**
  1. The key phrase, full employment level of income, was missing from the definition of deflationary gap.
  2. Showing deflationary gap at the underemployment equilibrium level, instead of at full employment level of income in the diagram.
  3. Improper/ incorrect/ no labelling of the diagram making it meaningless.
  4. Unnecessarily explaining fiscal measures to reduce the gap when only monetary measures were asked for.

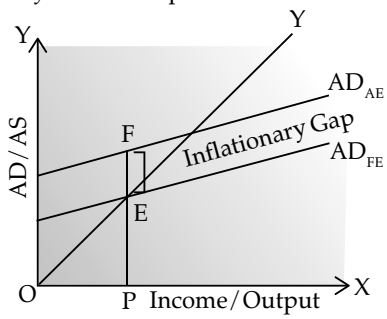
Answering Tips

- Learn the concept of deflationary gap with the help of a graph emphasizing the importance of the key word 'full employment'. Discuss the differences between inflationary and deflationary gap, preferably drawing two diagrams side by side, along with a columnar table with differences between the two written in point form. Students should give adequate practice to draw and label graphs correctly.
- Monetary measures and fiscal measures should be understood distinctively so that there is no confusion between the two.
- Students should be keep in mind the amount of content they ought to write for a 6-mark and a 3-mark question.

Q.2. Explain the concept of inflationary gap with the help of a diagram. Discuss two monetary measures to correct it. [ISC 2018]

Ans. Excess demand refers to a situation when aggregate demand exceeds aggregate supply corresponding to full employment.

When aggregate demand is higher than aggregate supply at full employment, the gap is called Inflationary Gap. That is, it occurs when Aggregate Demand is greater than Aggregate Supply (AD>AS) at the full employment level. In the inflationary gap situation, there is prevalence of inflation in the economy and so the price levels rise.



From the figure we can see that the AD_{FE} is the aggregate demand curve at the full employment level and AD_{AE} is the aggregate demand curve above full employment level. The point E is the equilibrium point where AD = AS.

But the excess demand (current) of ADAE, aggregate demand FP is more than the aggregate supply in the economy. This difference of actual aggregate demand and supply i.e., EF is the Inflationary Gap.

Inflationary Gap = Excess Demand = ADAE – ADFE = EF

The monetary measures to overcome the situation of inflationary gap are:

(i) Increase in Cash Reserve Ratio: An increase in Cash Reserve Ratio (CRR) may reduce the credit creation capacity of the commercial banks in the economy. This may lead to a fall in the borrowings from banks causing a fall in Aggregate Demand in the economy and helps to correct the inflationary gap in the economy.

(ii) Increase in Repo Rate: Repo rate is the rate at which the Central Bank lends money to the Commercial Banks. To correct the situation of Inflationary Gap, Repo Rate is increased. As a follow-up action, Commercial Banks raise the market rate of interest (the rate at which the Commercial Banks lends money to the consumers and the investors). This reduces demand for credit. Consequently, consumption expenditure and investment expenditure get reduced. Implying a reduction in Aggregate Demand, as required to correct Inflationary Gap.

(iii) Increase in Reverse Repo Rate: Reverse Repo Rate is the rate of interest at which commercial banks can park their surplus funds with the central bank, for a relatively shorter period of time. To deal with the situation of excess demand this rate may be increased by the central bank. It may encourage the commercial bank to park their surplus funds with

Central Bank. As a result, the availability of creating credit with the commercial bank will be reduced. Consequently, consumption expenditure and investment expenditure may get reduced, implying reduction in Aggregate Demand.

(iv) **Open Market Operations:** Sale of Government Securities in the open market by the Central Bank will adversely affect the ability of the Commercial Banks to create credit in the economy. As a result, Aggregate Demand in the economy may fall and correct the inflationary gap in the economy.

(v) **Margin Requirement:** Margin requirement is the difference between amount of loan offered and the market value of security offered against the loan. Higher the margin requirement lesser will be the demand for loan. In this situation of excess demand, there is a need to restrict loan/credit which can be done by raising the margin requirements.

[Any Two]

Examiner's Comment

- Several candidates were unable to show the inflationary gap at the full employment level of income for excess demand. In many answer scripts, the monetary measures were not explained correctly.

Answering Tips

- Learn the concept of inflationary gap and importance of full employment in the generation of this gap.
- The students should practice by drawing the diagram by showing the change in aggregate demand.

OBJECTIVE TYPE QUESTIONS

[A] Multiple Choice Questions

Q.1. Induced investment expenditure refers to: **[ISC 2023]**

(A) the investment made by a firm
(B) the expenditure, made by a firm, that remains constant
(C) the investment made in the economy that remains unaltered
(D) total investment expenditure which changes with change in national income

Ans. Option (D) is correct.

Explanation: Induced investment refers to that investment which changes as the level of income changes in the economy. It means Induced Investment is a capital investment that is influenced by National Income.

Q.2. If Marginal Propensity to Save (MPS) is 0.25 and initial change in investment is ₹250 crores, then the final change in income would be _____.

(A) ₹1,000 crores (B) ₹1,200 crores
(C) ₹500 crores (D) ₹3,500 crores

[OEB]

Ans. Option (A) is correct.

Explanation: $\Delta Y = \Delta I \times \frac{1}{MPS}$

$\Delta Y = ₹250 \times \frac{1}{0.25}$

Q.3. Income rises from ₹50,000 to ₹60,000, consumption increases from ₹40,000 to ₹48,000. In this situation, what will be the value of Marginal Propensity to Consume (MPC)?

(A) 0.80 (B) 0.20
(C) 0.10 (D) 0.90

[OEB]

Ans. Option (A) is correct.

Explanation: $MPC = \frac{\Delta C}{\Delta Y}$

$= 8000/10000$

$= 0.8$

Q.4. Suppose in a hypothetical economy, the income rises from ₹5,000 crore to ₹6,000 crore. As a result, the consumption expenditure rises from ₹4,000 crore to ₹4,600 crore. Marginal propensity to consume in such a case would be _____.

(A) 0.8 (B) 0.4
(C) 0.2 (D) 0.6

[OEB]

Ans. Option (D) is correct.

Explanation: $MPC = \frac{\Delta C}{\Delta Y}$

$= \frac{600}{1000}$

$= 0.6$

Q.5. In case of an under-employment equilibrium, which of the following alternatives is not true?

(A) Aggregate Demand is equal to Aggregate Supply.
(B) There exists excess production capacity in the economy.
(C) Resources are not fully and efficiently utilised.
(D) Resources are fully and efficiently utilised.

[OEB]

Ans. Option (D) is correct.

Explanation: When aggregate demand is equal to aggregate supply at less than full employment, it is a situation of under employment equilibrium.

Q.6. Equilibrium level of income/output is established when:

(A) AS = AD (B) C = I
(C) S = Y (D) none of the above

[OEB]

Ans. Option (A) is correct.

Explanation: Equilibrium output refers to a situation when in an economy: AD = AS. So that, all the producers who wish to produce during the year is exactly equal to what the buyers wish to spend on the purchase of goods and services during the year.

Q.7. If the marginal propensity to consume is greater than marginal propensity to save, the value of the multiplier will be:

(A) greater than 2 **(B)** less than 2
(C) equal to 2 **(D)** equal to 5

[OEB]

Ans. Option (A) is correct.

Explanation: MPC = 0.6

MPS = 0.4

$$K = \frac{1}{MPS}$$

$$= \frac{1}{1 - MPC}$$

$$= \frac{1}{0.4}$$

$$= 2.5$$

Q.8. The value of multiplier is:

(A) $\frac{1}{MPC}$ **(B)** $\frac{1}{MPS}$

(C) $\frac{1}{1 - MPS}$ **(D)** $\frac{1}{MPC - 1}$

[OEB]

Ans. Option (B) is correct.

Q.9. When aggregate demand is greater than aggregate supply, inventories:

(A) Fall **(B)** Rise
(C) do not change **(D)** first fall, then rise

[OEB]

Ans. Option (A) is correct.

Explanation: If aggregate demand is greater than aggregate supply, flow of goods and services in the economy tends to be less than their demand, which means the existing stock of producers would be sold out.

Q.10. A situation when AS = AD along with fuller utilisation of resources in the economy is called:

(A) underemployment equilibrium.
(B) inflationary gap.
(C) equilibrium without excess capacity.
(D) deflationary gap. **[OEB]**

Ans. Option (C) is correct.

Explanation: When AD = AS, all the producers wish to produce during the year is exactly equal to what the buyers wish to spend on the purchase of goods and services during the year, so, there is no excess capacity.

[B] Assertion Reason Questions

Directions: In the following questions, a statement of Assertion (A) is followed by a statement of Reason (R). Mark the correct choice as:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.1. Assertion (A): Aggregate Demand is expressed in terms of total expenditure made in the economy.

Reason (R): Aggregate Demand constitutes of Consumption Expenditure, Investment Expenditure, Government Expenditure and Net Exports. **[OEB]**

Ans. Option (A) is correct.

Explanation: Aggregate Demand refers to the value of final goods and services which all sectors of an economy are planning to buy during a year.

Q.2. Assertion (A): Full employment is the situation where all those workers who are able to work and willing to work get employment at the prevailing wage rate.

Reason (R): The situation of full employment is achieved only when the economy is in equilibrium. **[OEB]**

Ans. Option (C) is correct.

Explanation: Full employment equilibrium refers to a situation when equilibrium is attained i.e., aggregate demand is equal to aggregate supply at full employment level.

Q.3. Assertion (A): There is an inverse relationship between the value of marginal propensity to save and investment multiplier.

Reason (R): Saving is a leakage in the circular flow of income. Greater the saving, greater the leakage and lower the value of investment multiplier. **[OEB]**

Ans. Option (A) is correct.

Explanation: Higher the MPS, lower the multiplier and lower the MPS, higher the multiplier, as $K = \frac{1}{MPS}$.

Q.4. Assertion (A): Saving and investment are always equal.

Reason (R): Planned S and Planned I are equal only at equilibrium level. **[OEB]**

Ans. Option (D) is correct.

Explanation: Saving (S) and Investment (I) can be realised (ex-post) and planned (ex-ante). Planned S and Planned I are equal only at equilibrium level. Realised S and realised I are always equal.

Q.5. Assertion (A): The Government increases taxes in order to correct the problem of Excess Demand.

Reason (R): When taxes are increased it reduces the purchasing power of the public, thus reducing the Aggregate Demand. **[OEB]**

Ans. Option (A) is correct.

Explanation: During inflation, government should raise rates of all taxes especially on rich people because taxation withdraws purchasing power from the taxpayers and to that extent reduces effective demand.

Case Based Questions

[A] Multiple Choice Questions

I. Read the following report and answer the questions that follow:

Due to the Covid-19 situation, there has been a fall in the savings of the people, and they are spending the savings. This has led to the consumption to be done by the reduction of the savings. In order to control the situation, the government has announced various stimulus packages to revive the economy and providing employment through road and railway building projects. Still the desired savings or ex-ante savings that is required for the functioning of the economy has reduced. It is seen that the private consumption expenditure, private investment expenditure and ex-ante savings has reduced the aggregate demand in the economy. **[OEB]**

(a) Due to the Covid-19 situation what has happened:

(A) ex-ante savings has increased
(B) ex-post saving has increased
(C) ex-ante saving has reduced
(D) ex-post savings has reduced

Ans. Option (C) is correct.

Explanation: As income is either consumed or saved Hence, when there is rise in consumption expenditure, then ex-ante savings falls

(b) The fall in the savings is due to:

(A) Decrease in income
(B) Increase in consumption
(C) Increase in government expenditure
(D) Using savings for consumption

Ans. Option (D) is correct.

Explanation: The fall in the savings is due to the reason that rising consumption expenditure is forcing households to withdraw out of saving.

(c) Which of options did the government use to increase the savings and consumption in the economy:

(1) Giving stimulus packages
(2) Reducing investment in infrastructure
(3) Open Market Operations
(4) Providing employment
(5) Investing in the economy

Choose the correct option:

(A) 1 and 2 only **(B)** 1, 2 and 3
(C) 1 and 4 **(D)** 1, 4 and 5

Ans. Option (C) is correct.

Explanation: To increase the rate of saving and consumption in the economy, government has released various stimulus packages as well as provided employment.

(d) It is seen that the private consumption expenditure, private investment expenditure and ex-ante savings has reduced the _____ in the economy.

(A) aggregate demand **(B)** aggregate supply
(C) investment **(D)** savings

Ans. Option (A) is correct.

Explanation: Aggregate demand has been reduced in the economy due to fall in private consumption demand, fall in private investment demand and due to fall in savings.

(e) What happens when the aggregate demand falls short of the aggregate supply in the economy?

(A) fall in prices.
(B) increase in repo rate
(C) fall in inventory
(D) none of the above.

Ans. Option (A) is correct.

Explanation: When aggregate demand is lower than aggregate supply at full employment, the gap is called deflationary Gap. That is, it occurs when Aggregate Demand is smaller than Aggregate Supply (AD<AS) at the full employment level. In the deflationary gap situation, there is prevalence of deflation in the economy and so the price levels fall.

[B] Subjective Based Questions

I. An economy is in short-run equilibrium when the aggregate amount of output demanded is equal to the aggregate amount of output supplied. In the AD-AS model, you can find the short-run equilibrium by finding the point where AD intersects SRAS. The equilibrium consists of the equilibrium price level and the equilibrium output. A good practice is to think of the short-run equilibrium as "how much real GDP is this economy creating right now, and what is the CPI an economy has right now?"

In our analysis of markets, an economy in disequilibrium results in price adjusting until the market finds an equilibrium. The same general idea applies to a short-run macroeconomic equilibrium as well, but with a minor modification. If the amount of output demanded is greater than the amount of output produced, people chase after the limited goods available and drive up the price level. In response to the increase in the price level, producers create more goods and services. This continues until the amount of aggregate production equals the amount of aggregate demand. **[OEB]**

(a) What is the above article talking about?

Ans. The above article is talking about the short run equilibrium.

(b) As per Keynes, what are the two approaches by which the short-run equilibrium can be studied?

Ans. The two approaches are:

(i) Aggregate demand – Aggregate Supply Approach.
(ii) Investment – Savings Approach.

(c) According to the article when is the economy in the short run equilibrium?

Ans. According to the article, the economy is in the short-run equilibrium when the aggregate amount of output demanded is equal to the aggregate amount of output supplied.

(d) With the help of the information given in the article what would happen if the aggregate demand is less than the aggregate supply in the economy?

Ans. When the aggregate demand in the economy is less than the aggregate supply, it will create a situation of excess supply, leading to the prices in the economy to fall down drastically and creating deflationary situation in the economy. The government in such a case need to influx in cash with the help of increasing its expenditure.

(e) With the help of the information given in the article what would happen if the aggregate demand is more than the aggregate supply in the economy?

Ans. When the aggregate demand in the economy is more than the aggregate supply, it will create a situation of excess demand, leading to the prices in the economy to rise drastically and creating inflationary situation

in the economy. The government in such a case need to reduce the money supply in the economy with the help of decreasing its expenditure.

III. Central banks use tools such as interest rates to adjust the supply of money to keep the economy humming.

Monetary policy has lived under many guises. But however it may appear, it generally boils down to adjusting the supply of money in the economy to achieve some combination of inflation and output stabilization.

Most economists would agree that in the long run, output—usually measured by gross domestic product (GDP)—is fixed, so any changes in the money supply only cause prices to change. But in the short run, because prices and wages usually do not adjust immediately, changes in the money supply can affect the actual production of goods and services. This is why monetary policy—generally conducted by central banks such as the U.S. Federal Reserve (Fed) or the European Central Bank (ECB)—is a meaningful policy tool for achieving both inflation and growth objectives.

In a recession, for example, consumers stop spending as much as they used to; business production declines, leading firms to lay off workers and stop investing in new capacity; and foreign appetite for the country's exports may also fall. In short, there is a decline in overall, or aggregate, demand to which government can respond with a policy that leans against the direction in which the economy is headed. Monetary policy is often that countercyclical tool of choice. **[OEB]**

(a) What is the above article talking about? Explain in detail.

Ans. The above article is talking about Monetary Policy. Monetary policy is used by the central bank to control the inflation and deflation in the economy by the use of credit control measures.

(b) Name some of the tools that the central bank can use as its monetary policy?

Ans. The tools are:

(i) Repo rate
(ii) Reverse Repo Rate
(iii) Cash Reserve Ratio
(iv) Statutory Liquidity Ratio
(v) Open Market Operations

(c) How does Cash Reserve Ratio control inflationary pressures?

Ans. An increase in Cash Reserve Ratio (CRR) may reduce the credit creation capacity of the commercial banks in the economy. This may lead to a fall in the borrowings from banks causing a fall in Aggregate Demand in the economy and helps to correct the inflationary gap in the economy.

(d) How does reverse repo rate control deflationary pressure?

Ans. Reverse Repo Rate is the rate of interest at which commercial banks can park their surplus funds with the central bank, for a relatively shorter period of time. To deal with the situation of deficit demand this rate may be decreased by the central bank. It may discourage the commercial bank to park their surplus funds with central bank. As a result, the availability of creating credit with the commercial bank will be increase. Consequently, consumption expenditure and investment expenditure may get increased, implying reduction in deflationary pressure.

(e) How can the government use taxation policy to control both inflation and deflation?

Ans. (i) To curb the inflationary gap the government may increase the taxes. This may reduce the purchasing power in the hands of the public which in turn may reduce the Aggregate Demand in the economy to bring it equal to the Aggregate Supply.

(ii) To curb the deflationary gap the government may decrease the taxes. This may increase the purchasing power in the hands of the public which in turn may increase the Aggregate Demand in the economy to bring it equal to the Aggregate Supply.

Solutions for Practice Questions (Topic-1)

1 Mark Questions

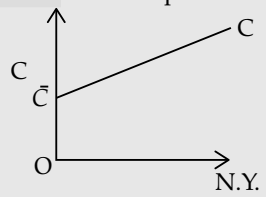
3. Aggregate supply can be defined as total output produced by all the producers of various goods and services in the economy.

2 Marks Questions

5. Propensity to save refers to different possible amounts of savings which the consumer desires to save at different levels of national income.

3 Marks Questions

6. Autonomous consumption is consumption when income is zero. It is independent of income.



$\bar{C}$ is the autonomous consumption.

[ISC Marking Scheme, 2015]

Examiner's Comment

- In some cases, consumption function was drawn as a horizontal straight line instead of positively sloped curve. Candidates could not relate income with consumption.

Answering Tip

- Explain to students that autonomous consumption is a part of aggregate consumption expenditure while the other part of it is a function of national income. Students should be told that consumption is independent of income.

Solutions for Practice Questions (Topic-2)

2 Marks Questions

5. We, know that MPC + MPS = 1

If, MPC = MPS,

Then, MPC + MPS = 1

$\Rightarrow$ MPS + MPS = 1

$\Rightarrow$ 2(MPS) = 1

$$MPS = \frac{1}{2} = 0.5$$

$$K = \frac{1}{MPS}$$

$$K = \frac{1}{0.5} = 2$$

4 Marks Questions

5. $$\text{Multiplier} = \frac{1}{1 - MPC}$$

When, MPC = 0.9

$$K = \frac{1}{1 - 0.9}$$

$$K = \frac{1}{0.1} = 10$$

$$K = \frac{\Delta Y}{\Delta I}$$

$$10 = \frac{\Delta Y}{32,000}$$

ΔY = ₹3,20,000 crore

Change in Final Income = ₹3,20,000 crore

Solutions for Practice Questions (Topic-3)

1 Mark Questions

5. Deficient demand can be defined as when aggregate demand is less than aggregate supply at the full employment level of income.

3 Marks Questions

5. Margin requirements can be defined as the difference between market value of security offered and the value of amount lent.

In case of deflation, margin requirements should be reduced so that the difference between the market value of security offered for loan and the amount of loan offered will get less.

Hence, this will lead to rise in money supply in the economy.

REFLECTION

(a) Can you with the help of graphs explain aggregate demand and its components?

(b) Can you calculate APC, APS, MPC and MPC?

(c) Do you understand what happens when either of aggregate demand or aggregate supply and Investment or savings are in disequilibrium?

(d) Can you calculate the investment multiplier and its components?

(e) Can you with the help of graph explain the situation of deflationary gap and inflationary gap?

(f) Do you understand the various monetary and fiscal policy tools that can be used to correct the situation of inflation and deflation?